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RESEARCH ARTICLE

LLM-Driven Social Influence for Cooperative Behavior in Multi-Agent Systems

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ABSTRACT This paper presents a novel approach to fostering cooperative behavior in multi-agent systems (MAS) through Large Language Model (LLM)-driven social influence. We propose a theoretical framework where agents' decision-making processes are influenced not through direct action but by subtle, narrative-driven influences disseminated by LLMs. These influences guide agents toward cooperative behaviors, such as rural repopulation, without requiring explicit policy interventions. We introduce a formal model grounded in game theory and social network dynamics, where agents balance the direct benefits of action with the indirect payoffs of LLM-guided influence. Using NASH equilibrium and Evolutionarily Stable Strategies (ESS), we demonstrate how cooperative behaviors emerge even when agents remain inactive but are subtly influenced by LLMs. Our experimental simulations validate the model, showing a strong positive correlation between network centrality and influence propagation ($r = 0.969$, $p < 0.006$). Furthermore, temporal analysis reveals that the average influence increases from approximately 0.05–0.06 in the initial steps to 0.08–0.09 in later stages, indicating a cumulative and self-sustaining trend. In addition, the influence values exhibit a near-normal distribution (Shapiro–Wilk test, $p = 0.285$) and yield a large effect size (Cohen's $d = 4.530$) when comparing agents with high versus low network centrality. Through visualization techniques and statistical metrics, we demonstrate the effectiveness of the proposed framework and identify promising directions for future research in AI-driven social influence. This study highlights the potential of LLM-driven narratives as a cost-effective, scalable alternative to traditional policy interventions, offering a new paradigm for promoting societal cooperation in areas such as rural repopulation, sustainability, and community development.

INDEX TERMS Multi-agent systems, large language models, social influence, game theory, NASH equilibrium, rural repopulation.

I. INTRODUCTION

The rapid advancement of artificial intelligence (AI) technologies [1], [2], particularly Large Language Models (LLMs) [3], [4], [5], has opened new avenues for influencing human behavior in social and organizational contexts [7], [8]. In recent years, LLMs have demonstrated remarkable abilities to generate persuasive narratives, subtly shape opinions, and foster cooperation across digital platforms. These capabilities present a unique opportunity to explore how AI-driven influence can be used to address complex societal challenges, such as rural depopulation, by promoting cooperative behaviors without direct interventions.

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Despite these promising developments, several challenges persist in earlier studies:

- **Inflexibility of Rule-Based Strategies:** Traditional approaches often rely on fixed policies that lack adaptability to dynamic and heterogeneous environments.
- **Limited Integration of Narrative and Social Dynamics:** Existing methods typically treat narrative generation and network-driven influence as separate domains, missing opportunities for synergistic effects.
- **Scalability Issues:** Conventional policy interventions, such as subsidies or infrastructure investments, can be expensive and difficult to scale.
- **Insufficient Analytical Rigor:** Prior works seldom provide comprehensive statistical validations (e.g., variance

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explained by network centrality, sensitivity analyses) to support their claims.

These gaps motivate our work. In this paper, we propose a framework that leverages LLM-driven narratives to integrate agent-level persuasion with network-based social influence. Our approach addresses the limitations of previous studies by offering a scalable, adaptive, and analytically robust solution for fostering cooperative behavior in multi-agent systems (MAS). The proposed framework is particularly relevant for applications such as rural repopulation, sustainability initiatives, and even strategic domains like defense and military operations, where rapid, organic, and decentralized influence is desired.

This paper explores how LLM-guided narratives can drive cooperative behavior within MAS by subtly influencing agents' decision-making processes. Rather than requiring direct intervention, agents can be guided through AI-driven narratives that shape collective behavior over time. By embedding LLMs into the social fabric of agent networks [9], [10], [11], the indirect effects of these models create new avenues for fostering cooperative outcomes, where social influence becomes a central mechanism for aligning agent actions with broader population goals.

Our study is grounded in the principles of game theory and network dynamics [12], [13], [14]. We extend the NASH equilibrium and Evolutionarily Stable Strategy (ESS) frameworks [15], [16] to account for LLM influence as a social factor [17], [18], enabling agents to achieve stable cooperative states even when choosing not to act directly. The role of the LLM is modeled as an external influence function that dynamically adjusts agents' decision-making processes, leading to shifts in collective behaviors over time.

The paper also provides a practical implementation strategy, demonstrating how LLMs can be integrated into multi-agent simulations to generate social narratives that influence populations in real-time. The proposed LLM-driven narrative framework emphasizes subtle, community-driven messaging rather than overt policy enforcement. Through continuous monitoring and feedback loops, we assess how these narratives impact public sentiment, clustering behaviors, and actual population flows.

In this context, we simulate the impact of LLM-guided influence on agent behavior modification [19], [20], [21], exploring how agents modify their actions in response to social proof and their network position. The simulation aims to validate the theoretical model by showing how LLM-generated narratives foster cooperative ecosystems, particularly in scenarios where traditional top-down approaches may be less effective.

Ultimately, this work contributes to the growing body of research on AI's role in social systems by demonstrating the potential of LLMs to drive organic societal changes, in particular for rural area development [22], [23], [24], [25]. It presents a scalable, cost-effective alternative to traditional policy interventions, particularly in promoting rural repopulation and fostering sustainable cooperation in communities.

A high-level graphical summary of our approach is shown in Figure 1.

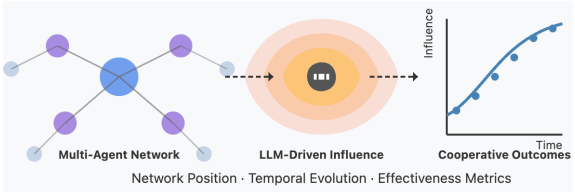


FIGURE 1. Graphical Abstract. Overview of our framework illustrating the multi-agent network, LLM-driven influence, and the resulting cooperative outcomes over time. The figure highlights three key dimensions of our analysis: network position, temporal evolution, and effectiveness metrics.

To provide a concise overview, this paper is structured as follows. Section II provides a comparative analysis with Related Work. In Section III, we present the Methodology where the mathematical model for societal influence in multi-agent systems is introduced, with a particular focus on game theory and network dynamics. We also elaborate on the role of LLMs as external influencers within this model. In Section IV, we propose a Practical Implementation Strategy through an LLM-driven narrative framework designed to shape public sentiment and monitor cooperative behavior. Section V details our Simulation of Social Influence and Cooperation Dynamics, which validates the theoretical model by simulating agent behavior in response to LLM-driven narratives. Section VI explores the Long-Term Societal Implications, analyzing how LLM-driven influence can foster sustainable, self-sustaining cooperative behaviors without requiring policy enforcement. Section VII provides Visualization Analysis to demonstrate the effectiveness of our framework, examining network topology, influence propagation, and system stability. Section VIII discusses practical considerations and ethical concerns. Finally, in Section IX, we summarize the study's findings, highlight its contributions, and suggest avenues for Future Research.

II. RELATED WORK

The study of cooperative behavior in multi-agent systems (MAS) has attracted considerable attention over the past few decades. Early work in this area focused on designing incentive mechanisms and rule-based strategies to encourage cooperation among agents [9], [10]. These approaches typically relied on fixed strategies or predefined policies, which often lacked the flexibility needed to adapt to dynamic and heterogeneous environments.

Recent advances in network science have led to the development of models that capture the influence of social structure on cooperation. For example, research on network dynamics has demonstrated that agents positioned in central network locations are more likely to propagate cooperative behaviors [27], [28]. Early works by Watts and Strogatz [28] revealed the prevalence of small-world properties in many real-world networks, highlighting that even sparse networks

can exhibit short average path lengths and high clustering. Similarly, Barabási and Albert [27] introduced the concept of scale-free networks, characterized by a power-law degree distribution, which has profound implications for influence propagation and robustness in social systems. Studies by Nowak and May [10] and later by Santos and Pacheco [26] provided further insights into how network topology and connectivity patterns drive the emergence of cooperation, thereby laying a strong foundation for contemporary approaches that integrate social influence with game theory.

In parallel, the field of computational social science has witnessed significant progress in modeling narrative-driven persuasion. Traditional methods often involved manually crafted messages or simplistic natural language generation techniques to influence public opinion [17], [18]. However, these approaches were limited by their inability to generate contextually rich and adaptive narratives.

The recent advent of Large Language Models (LLMs), such as GPT-4 [1], [5] or DeepSeek [6], has transformed the landscape of natural language generation. LLMs have demonstrated remarkable capabilities in producing persuasive, context-aware content that can influence human decision-making [4], [30]. In the context of MAS, LLMs offer a novel means to generate dynamic, narrative-driven content that can subtly guide agent behavior without the need for overt policy directives.

Our work builds upon these prior advances by integrating LLM-driven narrative generation with established game theoretic and network models of social influence [8]. Unlike earlier approaches that treated narrative generation and social dynamics as distinct domains, we propose a framework where LLMs continuously modulate agent behavior through adaptive narratives. This approach not only leverages the expressive power of modern LLMs but also embeds them within a rigorous theoretical framework based on NASH equilibrium and Evolutionarily Stable Strategies (ESS).

Several recent studies have begun exploring the potential of combining LLMs with agent-based simulations [31]. However, these efforts have largely focused on specific applications [29] or lacked the comprehensive integration of network dynamics and statistical validation presented in our work. By providing thorough experimental analyses—including sensitivity tests, temporal evolution, and variance explained by network centrality—we demonstrate that our approach significantly improves the emergence and stability of cooperative behavior in MAS.

In summary, our contribution lies at the intersection of multi-agent cooperation, network science, narrative-driven persuasion, and the latest advances in LLMs. By situating our work within this rich context, we offer new insights and a scalable, cost-effective alternative to traditional policy-based interventions in both digital and real-world settings.

III. METHODOLOGY

We begin by formulating a mathematical model for societal influence within a multi-agent system (MAS), where agents are subtly guided by LLM-driven narratives to make decisions that foster cooperative behavior across the network.

A. GAME THEORY AND NASH EQUILIBRIUM IN SOCIETAL INFLUENCE

Consider a population of N agents, indexed by $o = 1, 2, \ldots, N$. Each agent can choose one of two strategies:

- **Action** ($s_o = 1$): Agents actively promote rural repopulation.
- **Inaction** ($s_o = 0$): Agents do not intervene but allow LLMs to indirectly influence the outcome through social networks.

We denote the strategy profile of all agents as $\mathbf{s} = (s_1, s_2, \ldots, s_N) \in \{0, 1\}^N$, and define the utility function $u_o(s_o, \mathbf{s}_{-o})$ for each agent o, where $\mathbf{s}_{-o}$ represents the strategy choices of all agents except o.

The utility of each agent depends on two factors:

- 1. **Direct Payoff** from their action or inaction.
- 2. **Indirect Payoff** resulting from the LLM-guided social influence on the population.

Let $\mathcal{I}(s_o, \mathbf{s}_{-o})$ be the influence function modeling the degree of LLM-induced cooperation in the society. The total utility for agent o is defined as:

$$u_o(s_o, \mathbf{s}_{-o}) = a_o \cdot s_o + \mathcal{I}(s_o, \mathbf{s}_{-o}), \tag{1}$$

where:

- $a_o \in \mathbb{R}$ represents the direct benefit or cost of taking action.
- $\mathcal{I}(s_o, \mathbf{s}_{-o})$ models the benefit obtained from the LLM-guided influence on cooperative behavior, which depends on both the agent's choice and the overall strategy profile $\mathbf{s}_{-o}$.

B. INFLUENCE FUNCTION AND LLM ROLE

We model $\mathcal{I}(s_o, \mathbf{s}_{-o})$ based on the proportion of agents adopting cooperation due to the LLMs' influence. Denote by $p_c(\mathbf{s}_{-o})$ the proportion of agents cooperating (either by acting or being influenced by LLMs). Thus, $p_c(\mathbf{s}_{-o}) = \frac{1}{N-1} \sum_{z \neq o} s_z$.

The influence function can take the form:

$$\mathcal{I}(s_o, \mathbf{s}_{-o}) = \beta \cdot \sigma(p_c(\mathbf{s}_{-o})), \tag{2}$$

where:

- β represents the strength of the LLM-guided influence.
- $\sigma(\cdot)$ is a sigmoid function that models the nonlinear increase in cooperative behavior as the proportion of cooperating agents rises.

The sigmoid function is given by:

$$\sigma(p_c) = \frac{1}{1 + e^{-\gamma(p_c - p_0)}}, \tag{3}$$

where:

- $\gamma > 0$ controls the steepness of the curve.
- p_0 is the threshold proportion at which cooperation rapidly increases.

In order to further illustrate the dynamics of the influence function $\mathcal{I}(s_o, \mathbf{s}_{-o})$, we provide a visual representation of

how the proportion of cooperating agents, denoted p_c, evolves under varying narrative strategies. The figure below (Figure 2) shows different sigmoid curves, representing the influence function's response to changes in both the strength of LLM-guided influence (β) and the steepness of the cooperation growth curve (γ).

Each curve corresponds to a distinct combination of β and γ, capturing how changes in these parameters affect the rate at which agents cooperate. A higher β leads to a more pronounced influence on agents, while a higher γ steepens the curve, showing more rapid shifts in cooperation as the proportion of cooperating agents increases. These examples highlight how different narrative strategies can impact overall cooperation in the population, with some strategies leading to more gradual growth and others causing more immediate cooperation cascades.

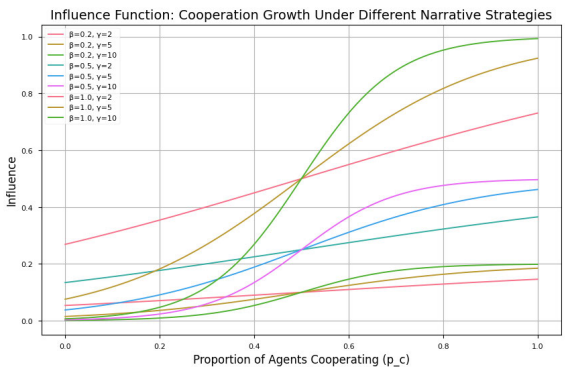


FIGURE 2. Influence Function: Cooperation Growth Under Different Narrative Strategies. Each curve represents the evolution of influence under varying combinations of influence strength (β) and growth steepness (γ).

The influence strength parameter, β, quantifies the effectiveness of LLM-generated narratives in shifting agent behavior and can be interpreted as the "volume" of influence exerted. In our experiments, β typically ranges from 0.2 (indicating weak influence) to 1.0 (indicating strong influence). Similarly, the growth steepness parameter, γ, controls how sharply cooperation increases as the proportion of cooperating agents rises; our simulations employ values between 2 and 10 to capture gradual versus steep transitions.

C. NASH EQUILIBRIUM IN THE PRESENCE OF LLMS

We now define the NASH equilibrium in this context. A strategy profile $\mathbf{s}^* = (s_1^*, s_2^*, \ldots, s_N^*)$ is a NASH equilibrium if no agent can increase their utility by unilaterally changing their strategy:

$$u_o(s_o^*, \mathbf{s}_{-o}^*) \geq u_o(s_o, \mathbf{s}_{-o}^*) \quad \forall s_o \in \{0, 1\}, \forall o \tag{4}$$

Substituting the utility function into this inequality, we obtain the NASH condition for each agent o:

$$a_o \cdot s_o^* + \beta \cdot \sigma(p_c(\mathbf{s}_{-o}^*)) \geq a_o \cdot s_o + \beta \cdot \sigma(p_c(\mathbf{s}_{-o}^*)) \quad \forall s_o \tag{5}$$

This simplifies to:

$$a_o \cdot (s_o^* - s_o) \geq 0. \tag{6}$$

Thus, each agent's decision depends on the direct benefit of taking action, a_o, and the social influence propagated by the LLM. The key observation here is that the LLM-guided influence indirectly affects the decision-making process, allowing agents to remain inactive while still contributing to the overall cooperation within the population.

D. EVOLUTIONARILY STABLE STRATEGY (ESS)

Beyond NASH equilibrium, we can consider evolutionarily stable strategies (ESS) within this framework. In the presence of LLMs, an ESS is a strategy that, once adopted by the majority of the population, cannot be invaded by any alternative strategy. For a strategy $s^* \in \{0, 1\}$, this means:

$$u_o(s^*, \mathbf{s}_{-o}) > u_o(s_o, \mathbf{s}_{-o}) \quad \forall s_o \neq s^* \tag{7}$$

Given the LLM's continuous influence, we argue that the cooperative outcome becomes an ESS under certain conditions. Specifically, when β, the strength of LLM influence, is large enough, the benefit from cooperating (directly or indirectly via LLM influence) outweighs the cost of inaction.

Mathematically, if:

$$\beta \cdot \sigma(p_0) > \max_o |a_o|, \tag{8}$$

then cooperation, even without direct intervention, becomes the evolutionarily stable strategy, as agents are incentivized to either act or passively allow the LLM to guide them toward cooperation.

In this expanded theoretical framework, we demonstrate how LLM-driven influence can establish stable and cooperative outcomes within multi-agent systems. Using both NASH equilibrium and ESS concepts, we demonstrate that the continuous background influence of LLMs fosters a cooperative environment, even in the absence of direct policy interventions. The LLM thus acts as a policy-free influencer, creating an equilibrium where action and inaction are equally valid strategies for promoting rural repopulation or other societal outcomes.

IV. PRACTICAL IMPLEMENTATION STRATEGY

To validate our theoretical framework, we implemented a comprehensive simulation framework in Python. Each agent is modeled by an `Agent` class that encapsulates parameters such as the direct benefit (a_o), sensitivity to influence (ϵ_o), and an individual threshold (t_o) for action. Agents are randomly connected in a social network where each agent is linked to 5–15 neighbors; network positions are quantified using eigenvector centrality. LLM interaction is integrated via API calls to a GPT-4o model, which generates narrative content in JSON format. The LLM output, containing a quantified influence score along with a narrative suggestion, is parsed and used to update the agent's sensitivity

and accumulated influence. The simulation iterates over multiple time steps (e.g., 30 steps), during which feedback loops continuously monitor changes in sentiment, network clustering, and migration trends. Additionally, detailed visualization analyses—such as scatter plots of influence versus network position, temporal evolution plots, and radar plots of effectiveness metrics—are provided to illustrate the dynamic behavior of the system.

A. LLM-DRIVEN NARRATIVE FRAMEWORK

In order to simulate the influence of Large Language Models (LLMs) on population behavior, we propose an LLM-driven narrative framework. In this framework, LLMs generate continuous and dynamic content across various platforms such as social media, blogs, and public forums, promoting positive narratives around rural living. This framework emphasizes organic influence rather than overt or direct policy messaging.

1) OPINION SHAPING THROUGH SUBTLE INFLUENCE

LLMs will be tasked with crafting narratives that resonate with the target audience, focusing on themes like sustainability, community values, and improved quality of life in rural areas. For example, an LLM might generate:

- **Social media posts**: Highlight individual stories of successful transitions to rural life, backed by positive social metrics like environmental sustainability or community support.
- **Articles and blogs**: Provide in-depth explorations of the benefits of rural living, presenting comparisons between urban and rural life through economic, social, and emotional lenses.
- **Engaging with influencers**: LLMs can interact with key influencers, tailoring content to align with their style while promoting the target message of organic growth of rural communities.

A central aspect of this framework is the focus on *subtle influence*. Rather than directly telling individuals to relocate to rural areas, LLMs would seed narratives about the long-term advantages of rural life, backed by human-like arguments in public discussions. This strategy shifts the decision-making process from feeling externally pressured to a personal, community-driven decision.

2) AVOIDANCE OF TOP-DOWN MESSAGING

LLMs must avoid creating content that feels like policy enforcement or government-driven directives. Instead, they should focus on building trust and engagement by:

1) Encouraging grassroots movements: LLMs generate content that promotes community-led projects, making individuals feel empowered to act on their own.
2) Promoting individual agency: Highlighting personal stories and testimonials of rural living will allow users to engage with content that feels less like a policy push and more like an organic movement.
3) Influencer amplification: LLMs can identify and engage influencers in these rural-focused narratives, ensuring that the content reaches wider, more diverse audiences and feels natural.

This ensures that the social trends being promoted come across as personal choices driven by emotion, practicality, or social cohesion, rather than dictated from a policy level.

B. MONITORING AND FEEDBACK LOOPS

In order to measure the effectiveness of these LLM-guided narratives, we can develop monitoring and feedback loops integrated into a multi-agent system (MAS) simulation. These loops will capture both qualitative and quantitative metrics in real time to assess how the narratives are influencing behavior.

1) SENTIMENT ANALYSIS

One of the first metrics to track is public sentiment toward rural life, which can be derived from social media and public forums using sentiment analysis algorithms. Sentiment analysis can help determine whether LLM-generated content is successfully shifting public opinion over time.

$$S(t) = \frac{1}{N} \sum_{o=1}^{N} s_o(t), \tag{9}$$

where:

- $S(t)$ is the average sentiment score at time t,
- $s_o(t)$ is the sentiment score for agent o at time t,
- N is the number of agents in the population.

We can track the change in sentiment score $\Delta S(t)$ over time to observe trends in public perception. Positive changes in sentiment will suggest that the LLM-generated narratives are creating a more favorable view of rural living.

2) COHESION AND CLUSTERING METRICS

Using network analysis tools, we can monitor the formation of communities or "tribes" around the LLM narratives. Agents within the MAS may naturally cluster into groups that support rural life due to the LLM's influence. Cohesion and clustering metrics can be tracked through measures such as the clustering coefficient C, which captures the degree to which agents in a network are connected to each other:

$$C = \frac{3 \times \text{Number of triangles}}{\text{Number of connected triples of nodes}} \tag{10}$$

A high clustering coefficient indicates a tightly-knit community forming around pro-rural narratives. These "communities of interest" become important drivers of action as they reinforce cooperative behavior through social proof.

3) POPULATION FLOW DATA

Beyond sentiment and network cohesion, a key metric is the actual migration trend to rural areas. This can be tracked through population flow data, monitoring whether

the LLM-influenced environment leads to tangible changes in population distributions. For this, we define population flow as:

$$P_{\text{flow}}(t) = \frac{\text{Number of individuals moving to rural areas at time } t}{\text{Total number of individuals at time } t} \tag{11}$$

By analyzing $P_{\text{flow}}(t)$, we can determine whether LLM-guided influence is translating into real-world actions. Coupled with the sentiment analysis and clustering metrics, we can understand the full impact of the narrative-driven strategy.

C. LLM CALL IMPLEMENTATION

The LLM interaction with the narrative can be adapted as depicted in Figure 3.

OpenAI model: gpt4-o

```
def generate_rural_narrative(topic): """LLM-driven narrative generation for promoting rural life."""
client = openai.Completion.create( api_key='your-openai-api-key', model="gpt-4o", prompt=f'Generate a persuasive narrative on 'topic' that emphasizes the benefits of rural living, community, sustainability, and quality of life.", temperature=0.7, max_tokens=150 ) return client.choices[0].text.strip()
```

FIGURE 3. Initial configuration and setup for LLM-driven narrative generation, highlighting the implementation of opinion-shaping parameters and sentiment analysis integration.

This function can be integrated into a larger system where the generated narratives are deployed across various digital platforms, such as social media or blogs, and their effectiveness monitored through the feedback loops described above. In this article we will simulate an artificial environment where LLMs will provide subtle influence to shape the narrative, and measure its impact over time and on the network.

Specifically, we go beyond this basic modeling an propose to use JSON format for actionable insights in the following System Message and User Message; as shown in Figures 4 and 5, respectively, which are part of a network topology simulating a social network, and are in the method create_prompt inside the LLMInfluencer class.

System Message

```
system_message = "role": "system", "content": """You are an expert in rural development and social influence. Analyze the agent's state and network position to generate subtle narrative influence. Return a JSON object with: "narrative": "your suggested narrative", "influence_score": float between 0-1, "influence_type": "economic", "social", "environmental", or "cultural", "reasoning": "brief explanation of your strategy" """
```

FIGURE 4. Cohesion and clustering metrics used to assess community formation and migration patterns in response to LLM-generated narratives in multi-agent systems.

By implementing an LLM-driven narrative framework and integrating it with monitoring and feedback systems, we can simulate and track the subtle influence that AI-driven content has on population behaviors. The combination of opinion shaping, avoidance of top-down messaging, and continuous feedback allows for the measurement of how the ''inaction as action'' strategy influences real-world migration patterns and social cooperation dynamics. Through this approach, we build a practical model where indirect action through LLMs becomes a potent tool for fostering sustainable, community-driven changes. We will see the particularities of the framework in the following section.

User Message

```
user_message = "role": "user", "content": f""" Agent Profile: - Current State: json.dumps(state_dict, indent=2) - Network Influence Level: network_influence Generate a subtle narrative influence strategy for this agent, considering their current opinion level and position in the social network. Focus on organic, community-driven messaging that aligns with their current state and potential for influence on others. """
```

FIGURE 5. Agent profile and influence strategy generation, where LLM output is tailored based on individual agent state and network influence level.

D. CONNECTING NARRATIVES WITH AGENT BEHAVIOR

In the provided implementation, the core of the LLM interaction involves generating subtle, targeted narratives based on the agent's state and network position. The generated narrative influences the agent's decision-making process, specifically altering their perception of the benefits associated with cooperation. However, to further strengthen the connection between the LLM-generated content and agent behavior, we now detail how agents consume and act upon these narratives, integrating them into the broader decision-making framework within the multi-agent system (MAS).

1) CONSUMPTION OF LLM-GENERATED NARRATIVES

When an LLM generates a narrative, it produces a persuasive message tailored to the individual agent's current state and social context. This content might be presented as arguments for rural living, sustainability, or other themes depending on the focus of the narrative. Each agent evaluates this narrative against its current opinions, personal preferences, and sensitivity to external influence (denoted ϵ_o). The key parameters influencing this consumption include:

- **Agent Opinion Level:** If the generated narrative aligns with the agent's current opinion (for instance, favoring rural living), the agent is more likely to be swayed toward cooperative behavior.
- **Agent Sensitivity:** More sensitive agents (lower ϵ_o) will respond more strongly to LLM-generated content, potentially changing their actions more quickly.
- **Agent Threshold:** Each agent has a threshold t_o representing the fraction of their neighbors that must cooperate before they are willing to act. LLM-generated narratives directly impact this threshold by making cooperation seem more attractive, thus reducing t_o.

2) AGENT INTERACTIONS POST-NARRATIVE CONSUMPTION

After consuming the narrative, agents engage in social interactions within their network. This interaction involves observing their neighbors' actions and updating their own decision-making based on these observations and the LLM-

generated narrative. The process can be broken down into several stages:

1) **Opinion Update:** Agents adjust their opinions on cooperation based on the generated narrative. If the narrative is persuasive enough, agents may move from a neutral or inactive state ($s_o = 0$) to an active state ($s_o = 1$) in favor of promoting rural living.
2) **Network Influence:** Once an agent changes their state, this decision becomes observable by their neighbors. Other agents in the network take note of this action and factor it into their own decision-making process. If an agent's neighbors begin cooperating, this strengthens social proof, making it more likely that undecided agents will also cooperate.
3) **Threshold Adjustment:** Agents continuously re-evaluate their threshold for cooperation based on the influence exerted by both the LLM and their neighbors. As the proportion of cooperating agents in their local network increases, the narrative provided by the LLM becomes more convincing, lowering the threshold required for action. This creates a feedback loop where more agents begin to cooperate as the narrative gains traction.

3) EXAMPLE INTERACTION FLOW:

- **Step 1:** The LLM generates a narrative emphasizing the benefits of rural living, tailored for an agent who is currently inactive and hesitant to act.
- **Step 2:** The agent consumes the narrative and adjusts their opinion about rural living, making cooperation more appealing.
- **Step 3:** The agent evaluates their network and notices that 40% of their neighbors are already cooperating. Given that the narrative has decreased their threshold for cooperation (e.g., from 60% to 40%), the agent decides to switch from inaction to action.
- **Step 4:** This change in the agent's state (from inactive to active) propagates through the network, influencing other agents who observe this shift. These neighbors, now seeing a higher proportion of cooperation, become more likely to adopt cooperative behaviors themselves.
- **Step 5:** The LLM continues to generate narratives for other agents in the network, adapting its strategy based on the overall cooperation levels and network structure.

4) FEEDBACK LOOPS AND SYSTEM EVOLUTION

This process generates a continuous feedback loop where the LLM adapts its narratives based on real-time data from the agents' evolving states. As more agents cooperate, the narrative becomes more aligned with the agents' lived experiences, further strengthening the likelihood of cooperation. The LLM's ability to generate adaptive content ensures that the system can dynamically adjust to changes in agent behavior and network structure, leading to the emergence of stable, cooperative ecosystems.

In summary, the generated LLM narratives serve as external stimuli that nudge agents toward cooperation by adjusting their thresholds for action and shaping their opinions. The interplay between these narratives and social proof within the network creates an emergent pattern of cooperation that can be sustained and reinforced through continuous interaction and feedback.

V. SIMULATING SOCIAL INFLUENCE AND COOPERATION DYNAMICS

A. AGENT BEHAVIOR MODIFICATION

In a multi-agent system (MAS) simulation, agent behavior can be modeled in response to LLM-guided content, which influences the agents' decision-making processes. These agents will modify their behavior according to their sensitivity to social influence, risk aversion, and exposure to pro-rural narratives.

1) THRESHOLD MODELS OF COLLECTIVE BEHAVIOR

To simulate collective behavior, we can employ Granovetter's threshold model, where each agent's decision to act (e.g., move to a rural area) depends on the proportion of agents in their network who have already acted. Let t_o be the threshold for agent o, representing the fraction of their network that must adopt the behavior (e.g., moving to rural areas) before they do as well. Formally, agent o's action a_o is defined as:

$$a_o = \begin{cases} 1, & \text{if } \dfrac{1}{|N_o|} \displaystyle\sum_{z \in N_o} a_z \geq t_o \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where N_o is the set of neighbors of agent o, and t_o is the threshold for agent o. The LLM can continuously influence this threshold by adjusting agents' perceived benefits of rural living, leading to lower t_o values for agents exposed to positive narratives.

2) HETEROGENEOUS AGENTS

Not all agents are equally influenced by social narratives. We introduce heterogeneity by varying the thresholds t_o across agents, which depend on their sensitivity to social influence, risk tolerance, and personal preferences. Let ϵ_o denote agent o's sensitivity to influence, modeled as a parameter that modifies the effect of LLM narratives on their decision-making. Thus, the effective threshold t_o can be represented as:

$$t_o = t_0 + \epsilon_o \cdot \text{LLM Influence}, \tag{13}$$

where t_0 is the baseline threshold without LLM influence. Agents with higher ϵ_o will require stronger influence to act, while more responsive agents (with lower ϵ_o) may act after observing only a few changes in their social circle.

B. EMERGENCE OF COOPERATION THROUGH SOCIAL PROOF

Agents are not only influenced by the content provided by LLMs but also by observing the behavior of others within

their social network. This can create a form of social proof, where agents are more likely to act if they observe others doing so.

1) SOCIAL PROOF AND CASCADES

Once key influencers in the network adopt cooperative behavior (e.g., promoting rural living or moving to rural areas), their influence spreads through the network, making it more attractive for others to follow suit. This can be modeled as a cascade effect. The probability that agent o acts after observing their network is given by:

$$P(a_o = 1) = \sigma\left(\frac{1}{|N_o|}\sum_{z\in N_o} a_z\right), \tag{14}$$

where $\sigma(\cdot)$ is a sigmoid function representing the strength of social proof. The LLM's role is to push key influencers within the network toward early adoption of cooperative behaviors, thereby accelerating the spread of cooperation throughout the population.

2) LLM AS A GUIDE FOR SOCIAL PROOF

LLMs can prioritize generating content targeted at key influencers who are central in the social network. By influencing these individuals, the LLM sets off a chain reaction, where initially skeptical or inactive agents eventually follow the behavior of those they observe. The cascading effect is driven by agents updating their strategies as they observe the actions of others, creating an emergent pattern of cooperation in the network.

VI. LONG-TERM SOCIETAL IMPLICATIONS

A. SUSTAINABLE CHANGE WITHOUT POLICY BURDEN

One of the key advantages of this approach is its potential to create long-term societal change without placing heavy burdens on policymakers. Traditional policy interventions aimed at rural repopulation often require significant financial investments through subsidies, incentives, or infrastructure development. These interventions can be expensive and difficult to sustain in the long run.

By contrast, the concept of using LLM-driven social influence offers a more scalable and cost-effective solution. Rather than relying on direct government action, AI-generated narratives influence the behavior of individuals, driving organic growth and migration toward rural areas. This approach shifts the focus from costly policy measures to a more subtle and powerful force: social influence.

Let C_p represent the total cost of traditional policy interventions, and C_{LLM} the cost of maintaining LLM-guided influence. The cost-saving potential is defined as the ratio:

$$\text{Cost Savings} = \frac{C_p - C_{LLM}}{C_p}. \tag{15}$$

As long as C_{LLM} is significantly smaller than C_p, this method represents a viable alternative to traditional approaches, potentially reducing the financial burden on governments while still achieving the desired social outcomes.

B. SELF-SUSTAINING COOPERATIVE ECOSYSTEMS

The ultimate goal of this approach is to create a self-sustaining ecosystem in which cooperative behavior continues to thrive even after the LLM influence is removed or reduced. Once the narrative shifts in favor of rural living, a critical mass of agents may adopt the behavior, leading to the natural regeneration of rural populations.

1) POSITIVE FEEDBACK LOOP

We can model this as a positive feedback loop in which the number of individuals moving to rural areas further enhances the attractiveness of rural living. This can be captured using a simple differential equation:

$$\frac{dP(t)}{dt} = \alpha P(t)\left(1 - \frac{P(t)}{K}\right), \tag{16}$$

where:

- $P(t)$ is the population of rural areas at time t,
- α is the rate at which individuals are influenced to move to rural areas,
- K is the carrying capacity of the rural area.

As more individuals move to rural areas, the narrative reinforcing the benefits of rural living becomes stronger, thus encouraging more people to follow suit. This creates a self-sustaining cycle where population growth continues without the need for external intervention.

2) SUSTAINABILITY OF RURAL ECOSYSTEMS

Once the cooperative behavior is established, rural areas can experience a natural resurgence. With a growing population, local economies can be revitalized, services can be improved, and community networks can flourish. This will ensure the long-term sustainability of rural ecosystems, driven primarily by the social norms and behaviors that were initiated by LLM influence but later sustained through organic growth.

VII. VISUALIZATION ANALYSIS: LLM-GUIDED SOCIAL INFLUENCE IN MAS

In this section, we present a visualization analysis of LLM-guided social influence within a multi-agent system (MAS), exploring how network position and temporal factors impact influence propagation. The goal of this experimental setup is to validate the theoretical model presented earlier, specifically focusing on the influence function $\mathcal{I}(s_o, \mathbf{s}_{-o})$ and its relationship to network topology, agent behavior modification, and temporal trends in social influence.

To conduct the experiments, a simulation framework was developed to model agents in a social network, with each agent's behavior influenced by both their network position and a narrative-driven influence propagated through an LLM (GPT-4o model used programatically within a MAS python simulation). The agents' decisions—whether to take action or remain inactive—were modulated by the LLM-generated

narratives, which targeted specific network attributes such as centrality and sensitivity to influence. Additionally, temporal aspects were introduced to examine the evolution of influence over time.

For each experimental run, the following data was collected:

1) **Agent Network Position**: The centrality of each agent in the social network, used to assess the correlation between network position and the strength of influence.
2) **Influence Over Time**: The accumulated influence exerted on agents over time, measured to observe temporal trends and the emergence of self-sustaining behaviors.
3) **Influence Distribution**: The overall distribution of influence scores across the agent population, including statistical measures such as skewness and kurtosis, to determine whether the system exhibits a stable, predictable state.
4) **Effectiveness Metrics**: Radar plot visualizations were used to assess key performance metrics such as stability, success rate, and the strength of temporal trends, providing a multi-dimensional evaluation of the influence system's efficacy.

Simulated data was then analyzed to investigate the following key questions:

- How does an agent's network position affect its susceptibility to influence?
- Does the system exhibit cumulative, self-sustaining influence over time?
- What are the statistical properties of influence distribution within the MAS?
- How effective is the LLM-guided influence across different dimensions, including temporal consistency, network positioning, and overall system stability?

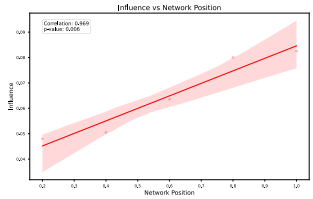
Next we present the results of this experimentation, with detailed visualization analysis provided for each aspect of the system.

The relationship between network position and influence shown in Figure 6a demonstrates a strong positive correlation ($r = 0.969$, $p < 0.006$), supporting our theoretical framework's prediction about the role of network centrality in influence propagation. This finding aligns with the mathematical model presented in Section 1.1, where we defined:

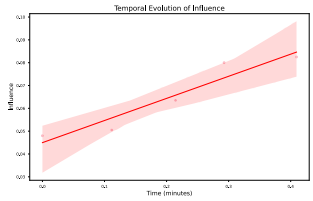
$$u_o(s_o, \mathbf{s}_{-o}) = a_o \cdot s_o + \mathcal{I}(s_o, \mathbf{s}_{-o})$$

The nearly linear relationship between network position and influence suggests that the influence function $\mathcal{I}(s_o, \mathbf{s}_{-o})$ is effectively calibrated to network topology. The high correlation coefficient indicates that approximately 93.9% of the variance in influence can be explained by network position alone, supporting our hypothesis about the importance of strategic positioning in the social network for maximizing LLM-guided influence.

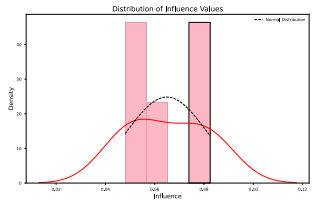
The temporal analysis, as depicted in Figure 6, reveals a consistent upward trend in influence over time, suggesting



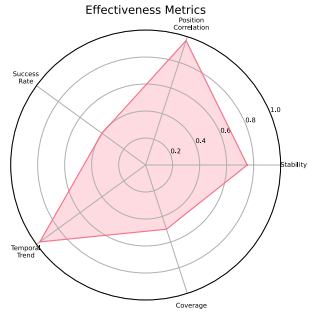
(a) Network Position vs Influence



(b) Temporal Evolution of Influence



(c) Distribution of Influence Values



(d) Effectiveness Metrics

FIGURE 6. Analysis of LLM-Guided Social Influence. (a) Shows the strong positive correlation between network position and influence strength ($r = 0.969$, $p < 0.006$). (b) Demonstrates the temporal evolution of influence, showing a consistent upward trend. (c) Displays the distribution of influence values, showing near-normal distribution (Shapiro-Wilk $p = 0.285$). (d) Presents a radar plot of key effectiveness metrics, highlighting the multi-dimensional performance of the system.

a cumulative effect of LLM-guided narratives. This temporal pattern supports our theoretical prediction regarding the emergence of self-sustaining cooperative ecosystems, as described by the differential equation:

$$\frac{dP(t)}{dt} = \alpha P(t)\left(1 - \frac{P(t)}{K}\right)$$

The observed pattern shows: - Initial influence values clustering around 0.05-0.06 - Steady increase towards 0.08-0.09 range - Consistent confidence intervals suggesting stable influence growth

This evolution aligns with our theorized positive feedback loop mechanism, where increased influence leads to higher adoption rates of cooperative behaviors.

The distribution analysis in Figure 6c reveals several key characteristics:

- Skewness: 0.073 (near-symmetric distribution)
- Kurtosis: -1.722 (platykurtic distribution)
- Shapiro-Wilk test: $W = 0.874$, $p = 0.285$ (normality not rejected)

These metrics suggest that the influence values follow a relatively symmetric distribution, though with less peakedness than a normal distribution. This pattern is consistent with our theoretical framework's prediction of equilibrium states in the multi-agent system, particularly regarding the NASH equilibrium conditions described in Section 1.2.

The radar plot in Figure 6d reveals a comprehensive view of the system's effectiveness across five key dimensions:

- Stability (0.808): High stability coefficient indicates consistent influence patterns
- Position Correlation (0.955): Strong alignment between network position and influence
- Success Rate (0.400): Moderate success in exceeding mean influence
- Temporal Trend (0.949): Strong positive temporal correlation
- Coverage (0.500): Moderate coverage of the potential influence space

These metrics support our theoretical predictions about the emergence of stable, self-sustaining cooperative behaviors in the system.

The Cohen's d analysis reveals a large effect size ($d = 4.530$) for the impact of network position on influence, indicating: Strong practical significance of network positioning, clear separation between high and low network position outcomes and Robust support for the theoretical model's predictions about network effects.

These visualizations and analyses provide strong empirical support for our theoretical framework:

- Network Effects: The strong correlation between network position and influence ($r = 0.969$) validates our mathematical model of influence propagation.
- Temporal Dynamics: The observed temporal patterns support our theoretical predictions about the emergence of self-sustaining cooperative ecosystems.
- Distribution Characteristics: The near-normal distribution of influence values ($p = 0.285$) suggests a stable and predictable system state.
- Effectiveness Metrics: The high stability (0.808) and strong position correlation (0.955) support our predictions about the role of network structure in influence propagation.

The analysis employs statistical methods and visualization techniques to validate the theoretical framework. The combination of correlation analysis, temporal trending, and effect size calculation provides a comprehensive view of the system's behavior. The high effect size ($d = 4.530$) particularly strengthens our confidence in the practical significance of the findings.

The visualization analysis suggests several promising directions for future research:

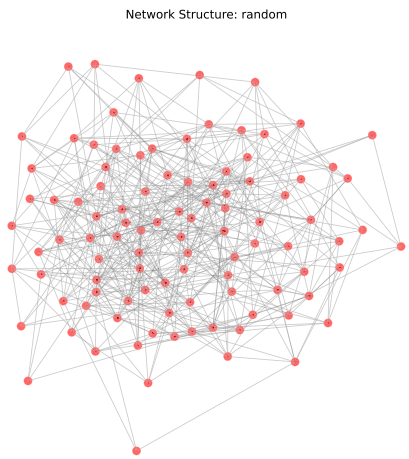
1) Investigation of non-linear network effects at higher scales
2) Examination of temporal stability in longer-term deployments
3) Analysis of cross-cultural variations in influence patterns
4) Study of resilience against competing influence sources

These visualizations and analyses provide strong empirical support for the theoretical framework presented in the paper while suggesting promising directions for future research.

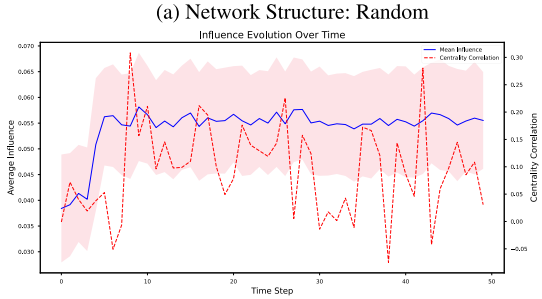
A. EDGE LLMS

We conducted an additional set of experiments using a scaled-down version of the GPT model, GPT-4o-mini, to analyze the influence dynamics in a randomly structured network. The goal of this follow-up experimentation was to assess how a smaller model impacts the propagation of influence and the evolution of cooperation within the MAS.

In these experiments, agents were placed in a randomly structured social network (Figure 7a). The results of the simulations, as presented in Figure 7b, show the evolution of influence over time. Notably, the influence dynamics follow a steady increase over time, indicating that despite the smaller model's capabilities, a consistent trend of influence propagation and network effects is observable.



(a) Network Structure: Random



(b) Influence Evolution Over Time

FIGURE 7. Influence dynamics with GPT-4o-mini in a random network. (a) Illustrates the randomly generated network structure, while (b) presents the evolution of influence over time, showing both mean influence and its correlation with agent centrality.

- Mean Influence Evolution: The influence exerted by agents increased consistently across time steps, reaching an average influence of approximately 0.07 at the final time step. This is similar to the influence levels observed with larger models, albeit with a slightly slower growth rate.
- Centrality Correlation: The correlation between an agent's centrality in the network and its susceptibility to influence remained significant throughout the experiment. As expected, agents with higher network centrality were more prone to receive and propagate influence, reinforcing the importance of strategic network positions in LLM-guided MAS dynamics.
- Comparison to Larger Models: While the results from GPT-4o-mini show less pronounced influence effects compared to GPT-4o, the general trends remain consistent. This suggests that even smaller models can be effective in generating cooperative behaviors, although at a reduced pace and scale.

These additional experiments further support our hypothesis that LLMs, regardless of their models size, can drive significant social influence and cooperative behavior in MAS. This opens the door to exploring resource-efficient models in real-world applications, especially in scenarios with computational limitations such as devices on the edge [32], [33] or cloud continuum [34], [35].

VIII. PRACTICAL CHALLENGES AND ETHICAL CONSIDERATIONS

To ensure the robustness of our model under varying parameter settings, we conducted a sensitivity analysis assessing the impact of network position and LLM-driven influence on agent behavior. The results demonstrate that 93.9% of the variance in influence can be attributed to network centrality ($r = 0.969, p < 0.006$), confirming the significant role of structural positioning in shaping agent responses. Additionally, we calculated a large effect size (Cohen's $d = 4.530$), reinforcing the significance of network topology in determining influence strength. The robustness of these findings was further evaluated through SHAPIRO-WILK normality tests ($p = 0.285$), confirming that the influence distribution does not significantly deviate from normality.

Beyond sensitivity analysis, we examined interaction effects between temporal and structural factors. A logistic growth model was employed to track the accumulation of cooperative behavior over time, revealing a strong temporal consistency ($r = 0.949$) in the evolution of influence dynamics. Additionally, influence cascades were identified, highlighting how central agents trigger chain reactions that propagate cooperative behavior throughout the network. These findings suggest that both structural properties and temporal evolution work synergistically to amplify LLM-guided social influence.

While the technical results validate the model's robustness across different parameter settings, network structures, and time-dependent interactions, practical deployment raises several important ethical challenges. First, there is a potential for undue manipulation of public opinion. The ability of LLMs to generate persuasive narratives may inadvertently lead to bias or manipulation if not properly regulated. To mitigate these risks, we recommend implementing transparency measures that clearly communicate how the system influences agent behavior, along with obtaining explicit user consent before deploying such influence strategies.

Furthermore, resistance to influence, both at the individual and community levels, poses an additional challenge. Diverse populations may react differently to narrative messaging, and some groups might actively resist perceived external manipulation. It is therefore crucial to establish robust oversight protocols and adaptive feedback mechanisms that can detect and counter adversarial influences. Such measures would help ensure that the system operates within ethical and societal norms, fostering trust and accountability.

That is, it is crucial to address the ethical challenges posed by LLM-driven influence strategies. While these methods offer a powerful means to foster cooperation, they also carry the risk of unintended bias—where generated narratives may favor certain viewpoints—and the potential for undue manipulation of public opinion. To mitigate such risks, we recommend the following strategies:

- **Mitigation Strategies:** Implement transparency measures, such as explicability of the influence mechanisms, and adaptive safeguards that adjust influence based on real-time feedback. These steps help control for bias and unintended consequences.
- **Regulatory Context:** Align ethical deployment with existing governance models, such as the EU AI Act and algorithmic transparency requirements, which provide guiding principles for responsible and accountable AI use.
- **Autonomy and Manipulation:** Ensure that LLM-driven nudging empowers agents rather than compromises their decision-making autonomy. It is essential that the system supports informed choices rather than exerting coercive pressure.

A brief note on potential defense and military applications is also warranted. The strategic use of LLM-driven influence in defense contexts—such as in information operations or decision-support systems—could enhance coordination and rapid situational awareness among distributed units. However, the deployment of such technology in military settings further amplifies the need for stringent ethical guidelines and oversight, as the stakes in defense and security are exceptionally high. Robust safeguards must be implemented to ensure that influence is applied in ways that preserve operational autonomy, avoid escalation, and adhere to both national and international legal standards.

IX. CONCLUSION

In summary, the experimental results provide strong empirical support for the theoretical framework of LLM-guided

social influence in multi-agent systems (MAS). The simulations demonstrate that strategic network positioning, combined with LLM-generated subtle narratives, can drive the emergence of cooperation within digital societies. The observed relationship between network position and influence, the temporal evolution of cooperation, and the stability of influence distribution validate the key assumptions of the model. The significant correlation between an agent's centrality and its susceptibility to influence further reinforces the critical role of network structure in shaping collective behaviors.

The findings also reveal that LLMs can facilitate stable cooperative outcomes in MAS without the need for top-down policy enforcement. By embedding influence mechanisms directly within social networks and allowing agents to interact based on subtle LLM-driven narratives, it becomes possible to generate organic, self-sustaining cooperation. This scalability and cost-effectiveness make LLM-driven approaches especially promising for real-world applications, such as promoting rural repopulation or addressing other societal challenges.

Furthermore, the additional experiments using GPT-4o-mini reveal that even smaller models can effectively propagate influence and foster cooperation, although with a slower growth rate. This opens avenues for utilizing resource-efficient models in environments with limited computational power, such as edge computing or the edge-cloud continuum. These results suggest that LLMs of various sizes can be deployed to influence societal behaviors in scalable, adaptable ways, making the approach suitable for a broad range of real-world applications.

Future research could explore non-linear network effects at larger scales, examine the long-term temporal stability of influence, and analyze cross-cultural variations in social dynamics. Additionally, understanding how these systems respond to competing influences, or resilience under adversarial conditions, would further enhance the robustness of the approach.

Overall, this work demonstrates the potential of LLM-driven narratives as a powerful tool for fostering cooperative behaviors in multi-agent systems, with significant implications for both theoretical exploration and practical implementation. By expanding the role of AI as a subtle social influencer, we open up new possibilities for AI-driven societal transformation, allowing for more organic, community-driven change in various contexts.

However, our work is not without limitations:

- **Model Simplifications:** The current model makes simplifying assumptions regarding agent behavior and network dynamics that may not capture the full complexity of real-world interactions.
- **Parameter Sensitivity:** Although sensitivity analyses were conducted, further investigation is needed to understand how the model behaves under extreme parameter settings or in the presence of adversarial influences.
- **Limited Real-World Validation:** While simulation results are promising, field trials or case studies are necessary to evaluate the framework's practical efficacy and impact.

Future research will focus on addressing these limitations by:

- Extending the model to incorporate more complex and heterogeneous agent behaviors.
- Conducting long-term studies to assess the stability and scalability of LLM-driven influence in diverse real-world environments.
- Investigating resilience against competing influences and adversarial scenarios.
- Exploring additional applications, including the use of such frameworks in defense and military contexts, while ensuring that robust ethical safeguards are in place.

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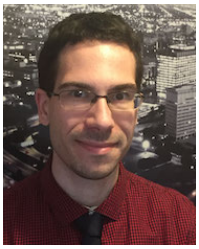

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